

A unique right factor from proportional oriented maximal minors

MathlibAnnex.Matrix.existsUnique_factor_of_orientedMaximalMinorsProportional

theorem

Recovers a square factor and its determinant from proportional determinants of every ordered row tuple.

Statement

Let A, B be I -by- n matrices over a field and let $\lambda \neq 0$. For an ordered tuple $\rho : \{1, \dots, n\} \rightarrow I$, let A_ρ, B_ρ be the selected square matrices, retaining the order and any repetitions in ρ . Assume

$$\det A_\rho = \lambda \det B_\rho \quad \text{for every } \rho.$$

Suppose one tuple ρ_0 satisfies $\det B_{\rho_0} \neq 0$. There is a unique square matrix L with $BL = A$ and $\det L = \lambda$.

Assumptions

The coefficient system is any field; I is any set, without an order or finiteness assumption. The scalar λ is nonzero, proportionality holds for all ordered tuples, and the same chosen tuple ρ_0 supplies the nonzero minor. The case $n = 0$ is included.

Conclusion

Writing $A_0 = A_{\rho_0}$ and $B_0 = B_{\rho_0}$, the factor is

$$L = B_0^{-1} A_0.$$

It satisfies $BL = A$ and $\det L = \lambda \neq 0$, hence is invertible. The uniqueness already follows from $BL = A$.

Notes

If I has a linear order, proportionality of the increasing-row minor families implies the ordered-tuple hypothesis: sorting an injective tuple introduces the same permutation sign for both matrices, while a repeated row makes both determinants zero. This adapter is separate from the theorem's order-free assumptions. A nonzero selected minor also implies that multiplication by B on column vectors is injective: restrict $Bv = 0$ to the selected rows and multiply by B_0^{-1} .

Proof route

Use the same nonzero selected minor to compare every row's coordinates by Cramer's rule. The coordinate equality supplies the global factorization; the selected square equation supplies its determinant and uniqueness.

▼ Proof steps

1. Proportionality at ρ_0 gives $\det A_0 = \lambda \det B_0$. Since λ and $\det B_0$ are nonzero in a field, $\det A_0 \neq 0$. Thus both A_0 and B_0 are invertible. Define $L = B_0^{-1}A_0$; immediately $B_0L = A_0$.
2. Fix any row index $i \in I$ and write a_i, b_i for the corresponding rows. For every position j , replace the j th entry of ρ_0 by i . The hypothesis applies even if this creates a repeated entry. The selected matrices are $A_0[j \leftarrow a_i]$ and $B_0[j \leftarrow b_i]$, so Cramer's row formula gives

$$\begin{aligned}(\det A_0)(a_i A_0^{-1})_j &= \det(A_0[j \leftarrow a_i]) \\ &= \lambda \det(B_0[j \leftarrow b_i]) \\ &= \lambda(\det B_0)(b_i B_0^{-1})_j = (\det A_0)(b_i B_0^{-1})_j.\end{aligned}$$

Cancellation of $\det A_0 \neq 0$ proves $a_i A_0^{-1} = b_i B_0^{-1}$.

3. Multiply the last row identity on the right by A_0 :

$$a_i = (a_i A_0^{-1})A_0 = (b_i B_0^{-1})A_0 = b_i L.$$

It holds for every i , hence $BL = A$. Taking determinants in the selected equation $B_0L = A_0$ gives $(\det B_0) \det L = \det A_0 = \lambda \det B_0$; canceling the nonzero factor $\det B_0$ yields $\det L = \lambda$.

4. If another matrix L' satisfies $BL' = A$, selecting the same rows gives $B_0L' = A_0$. Multiplication by B_0^{-1} forces $L' = B_0^{-1}A_0 = L$. This proves uniqueness even before imposing the determinant condition.
5. To pass from increasing minors to arbitrary tuples when an order on I is available, write an injective tuple as an increasing enumeration composed with a permutation σ . Both determinants gain the factor $\text{sgn}(\sigma)$, so their proportionality is unchanged. A noninjective tuple has two equal rows and both determinants vanish. For $n = 0$, there is one empty tuple, both determinants are 1, and the hypothesis reads $1 = \lambda \cdot 1$, forcing $\lambda = 1$; the unique empty matrix is the required factor.

Main citations

- Exact declaration and its source —
`MathlibAnnex.Matrix.existsUnique_factor_of_orientedMaximalMinorsProportional`
- Cramer's row-replacement coordinate formula —
`MathlibAnnex.Matrix.det_smul_vecMul_nonsingInv_eq_updateRowDet`
- A nonzero minor gives injectivity on column vectors —
`MathlibAnnex.Matrix.mulVec_injective_of_maximalMinor_ne_zero`
- An oriented minor from a row tuple — `MathlibAnnex.Matrix.orientedMaximalMinor`
- Proportionality for all ordered row tuples —
`MathlibAnnex.Matrix.OrientedMaximalMinorsProportional`
- Proportionality for increasing-row minors — `MathlibAnnex.Matrix.MaximalMinorsProportional`
- Agreement of increasing and oriented minors —
`MathlibAnnex.Matrix.orientedMaximalMinor_eq_of_orderEmbedding`
- The factor from the chosen square chart — `MathlibAnnex.Matrix.chartFactor`
- Changing a tuple entry replaces the selected row —
`MathlibAnnex.Matrix.submatrix_update_rowTuple`
- The selected matrices satisfy the factor equation —
`MathlibAnnex.Matrix.selectedSubmatrix_mul_chartFactor`

- The chart factor has the prescribed determinant — `MathlibAnnex.Matrix.det_chartFactor`
- Cramer comparison of arbitrary row coordinates —
`MathlibAnnex.Matrix.rowCoordinates_eq_of_orientedMaximalMinorsProportional`
- Row-coordinate comparison gives the global factor equation —
`MathlibAnnex.Matrix.mul_chartFactor_eq_of_orientedMaximalMinorsProportional`
- Uniqueness from the same chosen chart — `MathlibAnnex.Matrix.chartFactor_unique`
- A row permutation changes both minors by its sign —
`MathlibAnnex.Matrix.orientedMaximalMinor_permute`
- Repeated rows give a zero determinant —
`MathlibAnnex.Matrix.orientedMaximalMinor_eq_zero_of_not_injective`
- Sorting an injective row tuple — `MathlibAnnex.Matrix.exists_orderEmbedding_perm_of_injective`
- Transfer from increasing minors to oriented minors —
`MathlibAnnex.Matrix.orientedMaximalMinorsProportional_of_maximalMinorsProportional`
- The empty-dimensional scale is one —
`MathlibAnnex.Matrix.scale_eq_one_of_orientedMaximalMinorsProportional_zero`

Lean source signature (exact)

```
theorem existsUnique_factor_of_orientedMaximalMinorsProportional
  {n : ℕ} {ι : Type u} {K : Type v} [Field K]
  (source target : _root_.Matrix ι (Fin n) K) (scale : K)
  (hscale : scale ≠ 0)
  (hminor : OrientedMaximalMinorsProportional source target scale)
  (rows : Fin n → ι)
  (htarget : orientedMaximalMinor target rows ≠ 0) :
  ∃! L : _root_.Matrix (Fin n) (Fin n) K,
    target * L = source ∧ L.det = scale
```

In the source	Mathematical meaning
<code>{n : ℕ} {ι : Type u} {K : Type v} [Field K]</code>	$n \geq 0$, any row set I and any coefficient field. No order or finiteness of I is assumed.
<code>(source target : <i>root</i>.Matrix ι (Fin n) K)</code> <code>(scale : K)</code>	The two matrices A, B and scalar λ , respectively; thus <code>target * L = source</code> means $BL = A$.
<code>(hscale : scale ≠ 0)</code>	The hypothesis $\lambda \neq 0$.
<code>(hminor : OrientedMaximalMinorsProportional source target scale)</code>	For every ordered tuple $\rho : \{1, \dots, n\} \rightarrow I$, $\det A_\rho = \lambda \det B_\rho$. The cited definition retains the tuple order, including repetitions.
In the source	Mathematical meaning
<code>(rows : Fin n → ι) (htarget : orientedMaximalMinor target rows ≠ 0)</code>	The chosen tuple ρ_0 and the hypothesis $\det B_{\rho_0} \neq 0$. The same chosen rows form $A_0 = A_{\rho_0}$ and $B_0 = B_{\rho_0}$.

In the source	Mathematical meaning
$\exists! L : \text{root.Matrix } (\text{Fin } n) (\text{Fin } n) K, \text{ target } *$ $L =$ $\text{source} \wedge L.\text{det} = \text{scale}$	There is exactly one square matrix L satisfying both $BL = A$ and $\det L = \lambda$. Its formula in the conclusion is $L = B_0^{-1}A_0$. The case $n = 0$ is included.

▼ Exact surrounding binder context (separate excerpt)

```

noncomputable section

open scoped Matrix

namespace MathlibAnnex
namespace Matrix

universe u v

```

▼ Exact content identity

Declaration: MathlibAnnex.Matrix.existsUnique_factor_of_orientedMaximalMinorsProportional
Accepted content SHA-256: ee813a627f4b80001e5edf28439adb3034b15e6c5c1f0c8306a3e0164e66cf6e
Accepted source guide SHA-256: c465f63f209d9fa55fd71e5d2c19abf7e285f8574f16a90f907281ed6df239b4
Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73